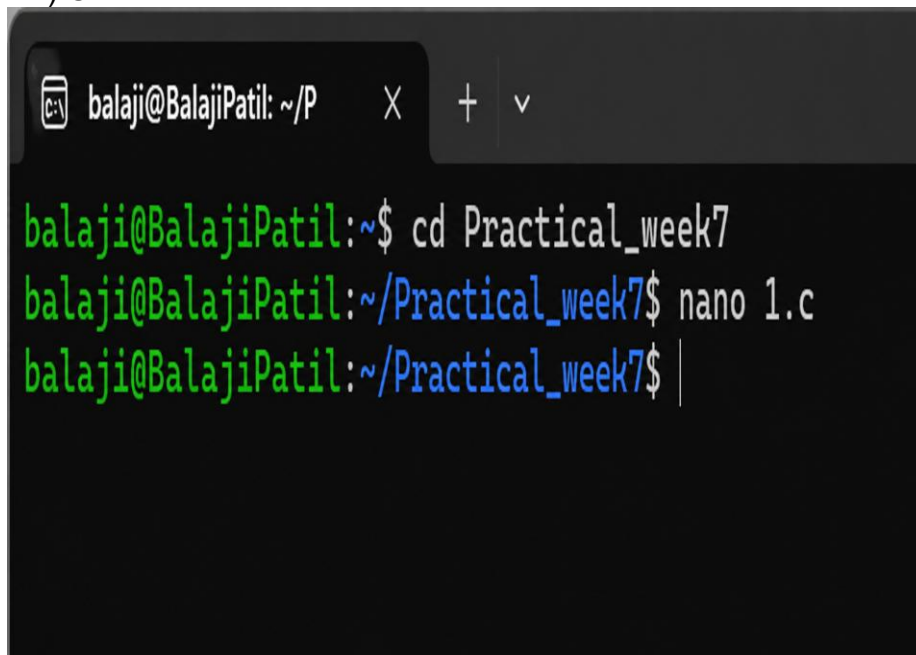

PRACTICAL WEEK 7

Task 1: Write a program that prints the addresses Of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

1) Create a file



```
balaji@BalajiPatil: ~/P X + v
balaji@BalajiPatil:~$ cd Practical_week7
balaji@BalajiPatil:~/Practical_week7$ nano 1.c
balaji@BalajiPatil:~/Practical_week7$ |
```

Then press enter to open command

2) write a code

```
GNU nano 8.7.1 1.c
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;           // Global variable
int global_uninit;           // Global uninitialized variable

void code_function()
{
    printf("Address of code function : %p\n", (void *)code_function);
}

int main()
{
    static int static_var = 20; // Static variable
    int stack_var = 30;         // Stack variable

    int *heap_var = malloc(sizeof(int));
    *heap_var = 40;

    printf("Linux Process Memory Address Layout\n\n");

    code_function();

    printf("Address of global variable      : %p\n", (void *)&global_var);
    printf("Address of global uninitialized : %p\n", (void *)&global_uninit);
    printf("Address of static variable       : %p\n", (void *)&static_var);
    printf("Address of heap variable          : %p\n", (void *)heap_var);
    printf("Address of stack variable          : %p\n", (void *)&stack_var);

    free(heap_var);

    return 0;
}
```

3)

- 1) To save press control O
- 2) To come back to shell press control X
- 3) Then DO compilation

```
balaji@BalajiPatil: ~/P
balaji@BalajiPatil:~/Practical_week7$ nano 1.c
balaji@BalajiPatil:~/Practical_week7$ gcc 1.c -o 1
balaji@BalajiPatil:~/Practical_week7$
```

OUTPUT:

```
balaji@BalajiPatil: ~/P  ×  +  ∨
balaji@BalajiPatil:~$ cd Practical_week7
balaji@BalajiPatil:~/Practical_week7$ nano 1.c
balaji@BalajiPatil:~/Practical_week7$ gcc 1.c -o 1
balaji@BalajiPatil:~/Practical_week7$ ./1
Linux Process Memory Address Layout

Address of code function      : 0x6056213561c9
Address of global variable   : 0x605621359010
Address of global uninitialized : 0x60562135901c
Address of static variable    : 0x605621359014
Address of heap variable      : 0x6056588e1010
Address of stack variable     : 0x7fff5cef318c
balaji@BalajiPatil:~/Practical_week7$ |
```

Task-2: Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
balaji@BalajiPatil:~$ cd Practical_week7
balaji@BalajiPatil:~/Practical_week7$
balaji@BalajiPatil:~/Practical_week7$ nano 2.d |
```

```
balaji@BalajiPatil: ~/P × + ∨
GNU nano 8.7.1 2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
int global_uninit;

int main()
{
    static int static_var = 20;
    int stack_var = 30;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 40;

    printf("Process ID: %d\n", getpid());
    printf("Code : %p\n", (void *)main);
    printf("Global : %p\n", (void *)&global_var);
    printf("Static: %p\n", (void *)&static_var);
    printf("Heap : %p\n", (void *)heap_var);
    printf("Stack : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");

    while (1)
    {
        sleep(1);
    }

    return 0;
}
```

OUTPUT:

```
balaji@BalajiPatil:~/Practical_week7$ nano 2.c
balaji@BalajiPatil:~/Practical_week7$ gcc 2.c -o 2
balaji@BalajiPatil:~/Practical_week7$ ./2
```

Process ID: 845

Code : 0x5c8d5ba951c9

Global : 0x5c8d5ba98010

Static : 0x5c8d5ba98014

Heap : 0x5c8d5ee44010

Stack : 0x7fffebc52aac

Process is running...

```
balaji@BalajiPatil: ~/P × + v
balaji@BalajiPatil:~$ cd Practical_week7
balaji@BalajiPatil:~/Practical_week7$ cat /proc/845/maps
5c8d5ba91000-5c8d5ba95000 r--p 00000000 08:30 52147 /home/balaji/Practical_week7/2
5c8d5ba95000-5c8d5ba96000 r-xp 00001000 08:30 52147 /home/balaji/Practical_week7/2
5c8d5ba96000-5c8d5ba97000 r--p 00002000 08:30 52147 /home/balaji/Practical_week7/2
5c8d5ba97000-5c8d5ba98000 rw-p 00002000 08:30 52147 /home/balaji/Practical_week7/2
5c8d5ba98000-5c8d5ba9a000 rw-p 00003000 08:30 52147 /home/balaji/Practical_week7/2
5c8d5ee44000-5c8d5ee65000 rw-p 00000000 00:00 0 [heap]
7e80eae00000-7e80eae28000 r--p 00000000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e80eae28000-7e80eafc0000 r-xp 00028000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e80eafc0000-7e80eb00e000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e80eb00e000-7e80eb012000 rw-p 0020d000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e80eb012000-7e80eb014000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7e80eb014000-7e80eb021000 rw-p 00000000 00:00 0
7e80eb114000-7e80eb120000 rw-p 00000000 00:00 0
7e80eb126000-7e80eb128000 rw-p 00000000 00:00 0
7e80eb128000-7e80eb12c000 rw-p 00000000 00:00 0
7e80eb12e000-7e80eb12e000 rw-p 00000000 00:00 0
7e80eb12e000-7e80eb13e000 rw-p 00000000 00:00 0
7e80eb130000-7e80eb131000 r--p 00000000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e80eb131000-7e80eb16b000 r-xp 00001000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e80eb160000-7e80eb16b000 r--p 00030000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e80eb16b000-7e80eb16d000 r--p 0003b000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e80eb16d000-7e80eb16e000 rw-p 0003d000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7e80eb16e000-7e80eb16f000 rw-p 00000000 00:00 0 [vvar]
7e80eb16f000-7e80eb171000 r--p 00000000 00:00 0 [vvar_vclock]
7e80eb171000-7e80eb173000 r-xp 00000000 00:00 0 [vdso]
7fffebc33000-7fffebc54000 rw-p 00000000 00:00 0 [stack]
balaji@BalajiPatil:~/Practical_week7$ |
```